Ex. No. : **10** **Date:**

Register No : **231701007** **Name: Badrish R S**

A python program to implement dimentionality reduction - PCA

Aim:
To implement a python program that implement dimentionality reduction - PCA

Requirements:

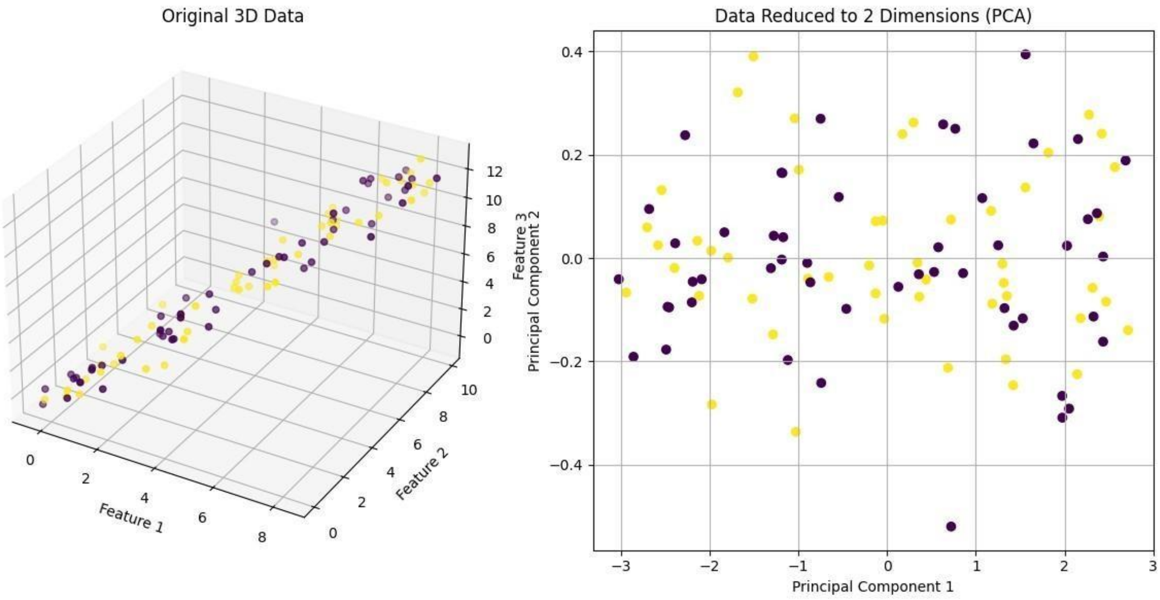
Jupyter Notebook with Python.

PROGRAM:

```
import numpy as np import matplotlib.pyplot as plt

from sklearn.decomposition import PCA from

sklearn.preprocessing import StandardScaler


print("--- Principal Component Analysis (PCA) Example ---")


np.random.seed(42)

X_3d = np.random.rand(100, 3) * 10

X_3d[:, 0] = X_3d[:, 1] * 0.8 + np.random.randn(100) * 0.5 X_3d[:, 2]

= X_3d[:, 1] * 1.2 + np.random.randn(100) * 0.5 y_labels =

np.random.randint(0, 2, 100)


print(f"Original data shape: {X_3d.shape}")
```

```
scaler = StandardScaler()

X_3d_scaled = scaler.fit_transform(X_3d)


n_components = 2 pca =

PCA(n_components=n_components)

X_2d = pca.fit_transform(X_3d_scaled)


print(f"Reduced data shape: {X_2d.shape}") print(f"Explained variance ratio by each component:

{pca.explained_variance_ratio_}") print(f"Total explained variance:

{pca.explained_variance_ratio_.sum():.2f}")


fig = plt.figure(figsize=(12, 6)) ax = fig.add_subplot(121,

projection='3d') ax.scatter(X_3d[:, 0], X_3d[:, 1], X_3d[:, 2],

c=y_labels, cmap='viridis') ax.set_title('Original 3D Data')

ax.set_xlabel('Feature 1') ax.set_ylabel('Feature 2')

ax.set_zlabel('Feature 3')


ax_2d = fig.add_subplot(122) ax_2d.scatter(X_2d[:, 0], X_2d[:, 1],

c=y_labels, cmap='viridis') ax_2d.set_title(f'Data Reduced to

{n_components} Dimensions (PCA)') ax_2d.set_xlabel('Principal

Component 1') ax_2d.set_ylabel('Principal Component 2')

ax_2d.grid(True)
```

```
plt.tight_layout() plt.show()
```



Output:

--- Principal Component Analysis (PCA) Example ---

Original data shape: (100, 3)

Reduced data shape: (100, 2)

Explained variance ratio by each component: [0.98827418 0.00918499] Total explained variance: 1.00

Result:

The python program has been successfully implemented